

Monthly Intelligence Report

Edition 01 — The North-East Estonia Corridor

Where the energy transition is outrunning the grid. Pre-launch · July 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

614

165 HV ≥110 kV · 37 MV (20-110 kV)
· 412 distribution-tier

GRID LINES MAPPED

5,355 km

3,984 HV ≥110 kV · 147 MV · OSM
live extraction

MC SIMULATIONS

39.8 M

20,000 per substation · Gaussian
copula

PUBLIC DATA SOURCES

28

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 130,000+ source data points per run, executes 39.8 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 0.91 billion arithmetic operations — producing 119,400 output data points with full uncertainty quantification across 614 substations (LIVE — OSM) and 4,757 grid lines spanning 5,355 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 28 verified public sources, making the methodology fully auditable and reproducible. Initially covering Estonia's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation EE_00307

Tartu, Tartu · 0 kV

0.418

R_FINAL

Medium

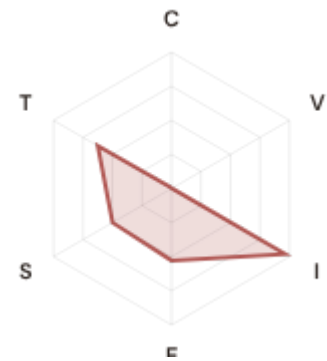
CLASSIFICATION

0.075

CI WIDTH

44th

FLEET PERCENTILE



COMPONENTS

- C** Continuity: 0.005 / 0.30 (2%)
- I** Infrastructure: 0.964 / 0.25 (385%) ▲
- S** Saturation: 0.500 / 0.20 (250%)
- V** Voltage: 0.004 / 0.10 (4%)
- E** Economic: 0.529 / 0.10 (529%)
- T** Transition: 0.625 / 0.05 (1250%)

MODIFIERS

- R3 Consequence:** 1.028 ↑ amplifies
- R4 Graph Criticality:** 1.000 → neutral
- R6a Restoration:** 1.000 → neutral
- R6b Seismic:** 1.002 → neutral
- R7 Digital Readiness:** 1.019 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Substation EE_00307** in Tartu, Tartu. With an R_final of 0.418 (Medium band, 44th fleet percentile), its risk profile is dominated by the T (Transition) component at 1250% of its weight ceiling, followed by E (Economic) at 529%. Modifiers collectively shift R_base by 7.6%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Estonia Numérique / Estonia 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0205

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

154

25.1% · R3 ≥ 1.08

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

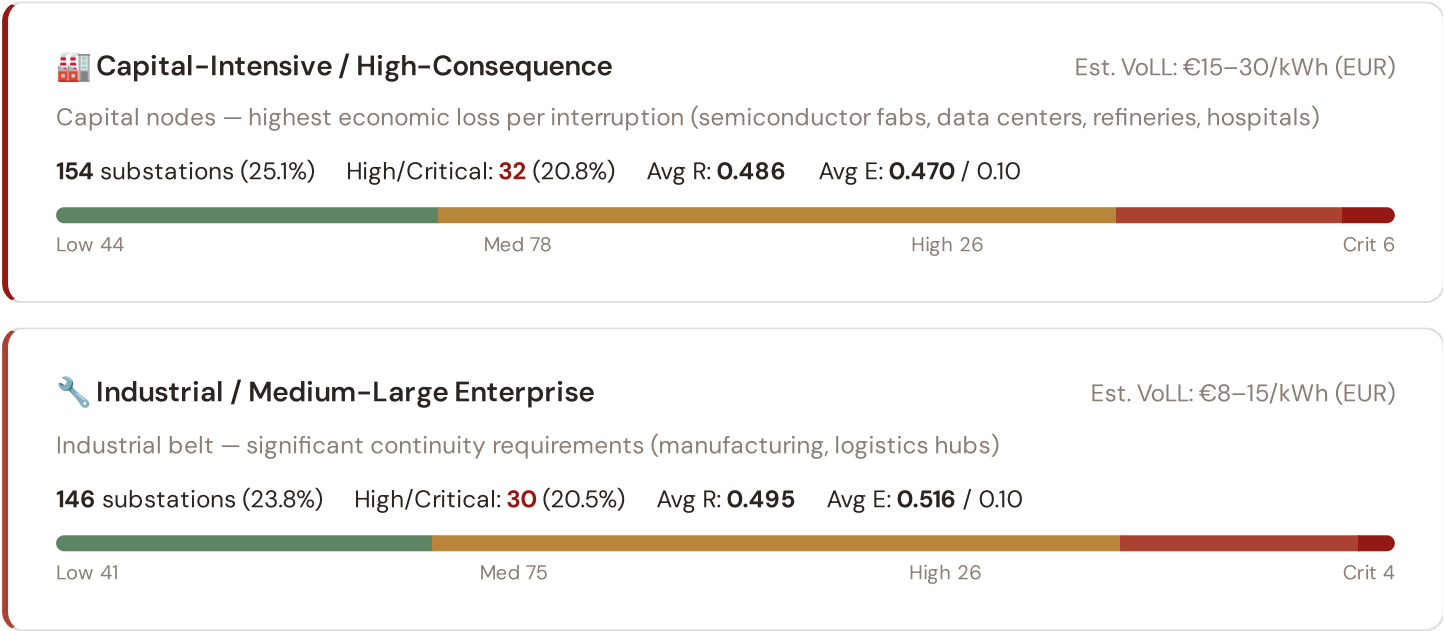


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0192$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in —. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 449 (73.1%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.





Commercial / SME-Dense

Est. VoLL: €3–8/kWh (EUR)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

160 substations (26.1%) High/Critical: **26** (16.3%) Avg R: **0.459** Avg E: **0.492** / 0.10



Light / Agricultural / Rural

Est. VoLL: €1–3/kWh (EUR)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

154 substations (25.1%) High/Critical: **14** (9.1%) Avg R: **0.382** Avg E: **0.514** / 0.10



Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **32 substations** combine capital-intensive economic fabric ($R3 \geq 1.08$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Ida-Viru (43)**, **Jõgeva (36)**, **Järva (15)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 6.2%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Maakond Digital Readiness Ranking

Maakond ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify Maakond combining digital exposure with high grid stress.

R7 Digital Readiness
Low High



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Hiiu		1.0192
2	Jõgeva		1.0198
3	Rapla		1.0199
4	Lääne-Viru Δ high R		1.0200
5	Põlva		1.0201
6	Tartu		1.0204
7	Ida-Viru Δ high R		1.0204
8	Järva Δ high R		1.0204
9	Viljandi Δ high R		1.0207
10	Pärnu Δ high R		1.0208
11	Valga Δ high R		1.0209

12	Lääne ▲ high R		1.0211
13	Harju		1.0214
14	Saare ▲ high R		1.0221
15	Võru ▲ high R		1.0221

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Hiiu** has the lowest average R7 (1.0192), while **Võru** leads at 1.0221 — a spread of 0.0029 across the R7 range. **0 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0205, median = 1.0192; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **28 verified public data sources** feeding 95 variables across 6 components and 6 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI’s credibility.

TOTAL SOURCES

28

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

11

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

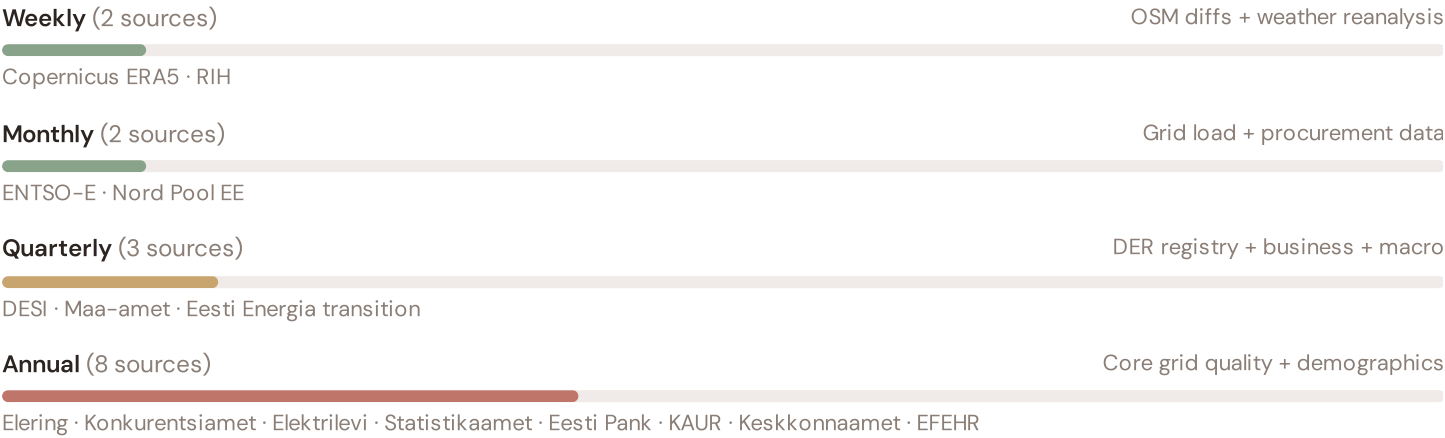
Data Source Registry — June 2026

Copernicus ERA5	WEEKLY	0.25°	12 vars
OpenStreetMap Overpass	CONTINUOUS	Per sub	8 vars
Eesti Energia phase-out tracker	QUARTERLY	Ida-Viru	5 vars
Konkurentsiamet	ANNUAL	Per-DSO	12 vars
Elektrilevi OÜ	ANNUAL	Maakond	15 vars
Eesti Pank	ANNUAL	National	8 vars
DESI 2024	ANNUAL	National	3 vars
Eurostat NUTS-3	ANNUAL	NUTS-3	18 vars
d09_brell_legacy (regime_change_2025_02)	ANNUAL	Sync area	6 vars
Keskkonnaamet EU-ETS	ANNUAL	Per plant	4 vars
ENISA cyber index	ANNUAL	National	3 vars
Tartu Univ. / Genome Center	ANNUAL	Tartu	3 vars
CEER Benchmarking 2024	ANNUAL	National	5 vars
TalTech	ANNUAL	Harju	2 vars
Tartu Observatory	ANNUAL	National	3 vars
IEEE C57.91 + CIGRE TB 761	STATIC	Eq. class	2 vars
Elering AS	ANNUAL + 15-MIN	TSO-wide	18 vars
Statistikaamet (Statistics Estonia)	ANNUAL + Q	NUTS-3	22 vars
Riigi Ilmateenistus (RIH)	DAILY + 30Y	Stations	14 vars
Maa-amet (Estonian Land Board)	BIENNIAL	Maakond	6 vars
Keskkonnaagentuur (KAUR)	BIENNIAL	National	5 vars
EFEHR ESHM20	5-YR UPDATE	Pan-EU	4 vars

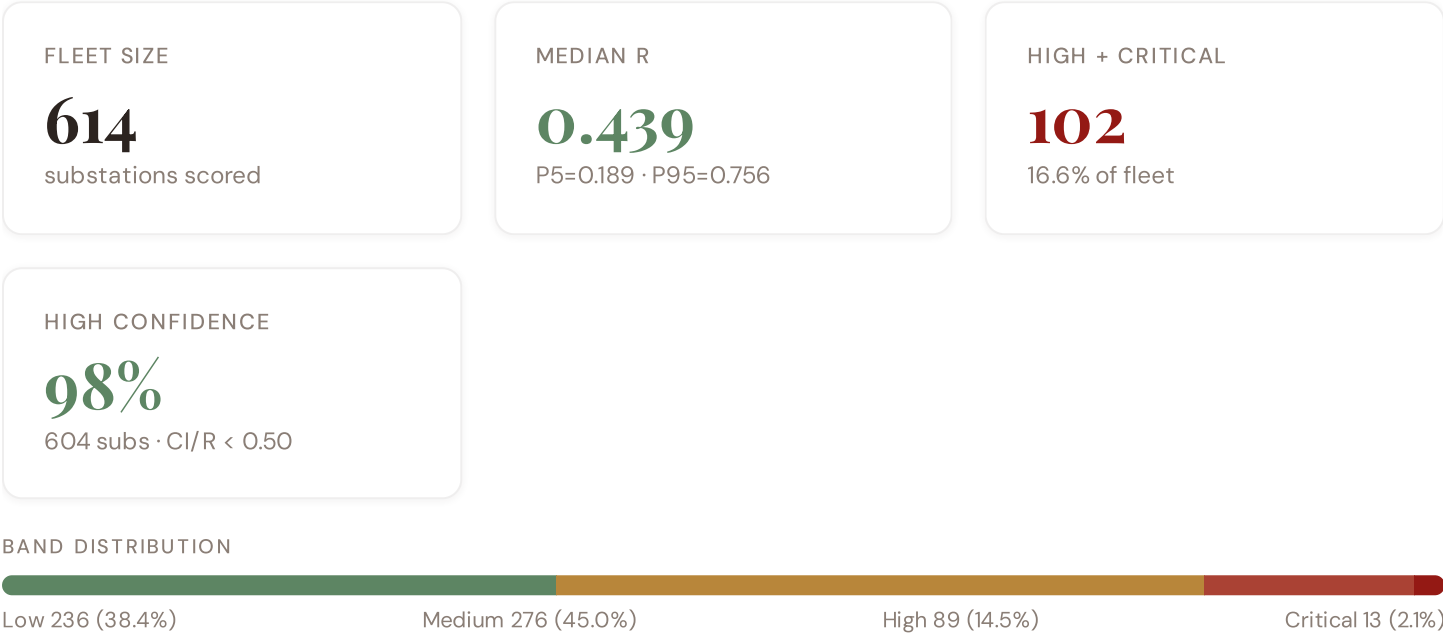
ENTSO-E Transparency	HOURLY	Zone	14 vars
Nord Pool EE bidding zone	HOURLY	EE zone	9 vars
d08_nordpool shared module	HOURLY	Baltic	5 vars
EU Just Transition Fund	MULTI-YR	Ida-Viru	4 vars
EstLink HVDC	HOURLY	EE-FI	4 vars
LitPol Link HVDC	HOURLY	LT-PL	4 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **614 substation scores remain unchanged** this edition. The fleet median R of 0.439 (mean 0.455) reflects a distribution skewed toward the lower bands, with 38.4% in Low and 45.0% in Medium. The first material score changes are expected when the national regulator

publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

9 changes tracked across PO, PI, and P2 releases

EE-S15-9	NEW	Hotfix #3 — R3 Light/Rural tier distinct (1.08, was collapsing to Commercial 1.06)
EE-S15-8	DATA	Hotfix #2 — substation socio_economic (GDP/cap, R&D, unemployment) + regions median_R alias
EE-S15-7	NEW	Hotfix #1 — grid-geo.json to FR-compatible schema + landing path active (red)
EE-S15-6	NEW	Wave D — 5-commit deploy to ikengassiindex.github.io/estonia/
EE-S15-5	NEW	Wave C — 8 HTML pages + ssi-metadata.js + grid-geo.json + bounds.json (geoBoundaries ADM1)
EE-S15-4	NEW	Wave B — scoring-ee engine: Markov (medium tier) + MC + Sobol + 614-sub R-distribution
EE-S15-3	DATA	Wave A — digital-twin-ee + 12 modules incl. d08_nordpool + d09_brell_legacy (NEW shared)
EE-S15-2	DATA	d05_osm LIVE — 1,949 raw → 614 substantive via KB §38 small-country filter (31.5% retention)
EE-S15-1	ENHANCED	KB v12 → v13 + BPG v1.11 → v1.12 — Part XIV Baltic Onboarding Roadmap

SECTION D — MONTHLY DEEP-DIVE

The North-East Estonia Corridor: Oil-Shale Phase-Out Meets BRELL Desync

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** NE-stressed cluster · **002** DER saturation hotspots (border maakonds) · **003** Climate vulnerability under CMIP6 · **004** Urban-Rural economic impact disparity · **005** coal-fleet decommissioning risk and transition risk · **006** Infrastructure age vs smart-meter rollout · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Maakond Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the NE-stressed cluster, Why Now



12
1 EHV · 11 HV

33.3% of corridor

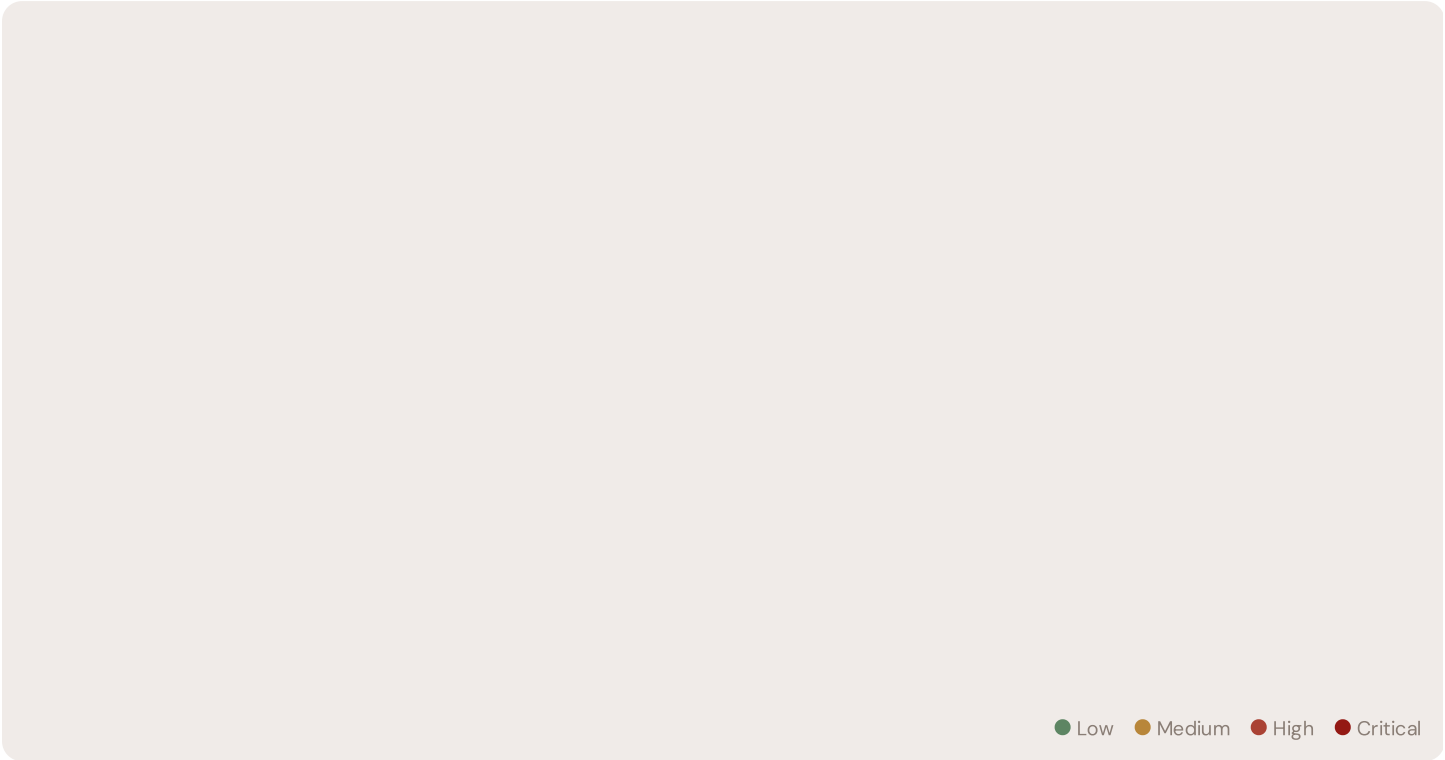
Fleet median: 0.439

WORST MAAKOND

Valga
Avg R: 0.560 · 12 substations

The Valga hosts **12 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **3 substations (25.0%)** are classified High and **1 Critical**, with only **3** in the Low band. 67% of corridor substations score above the national fleet median of 0.439. The corridor median R of 0.514 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	MAAKOND	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Loading...										

D.3 Component Analysis

The dominant risk driver across the Valga corridor is **T (Transition)**, averaging 0.703 against a fleet average of 0.540 — occupying 1406% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.593 vs fleet 0.498 (593% of

ceiling). Among modifiers, the R3 consequence multiplier averages 1.077 (fleet: 1.062), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.002, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Maakond Breakdown

The region regions-level breakdown reveals significant intra-regional variation. **Valga** leads with a mean R of 0.560 across 12 substations, while **Valga** scores 0.560 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Valga is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

4 Valga substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Valga sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

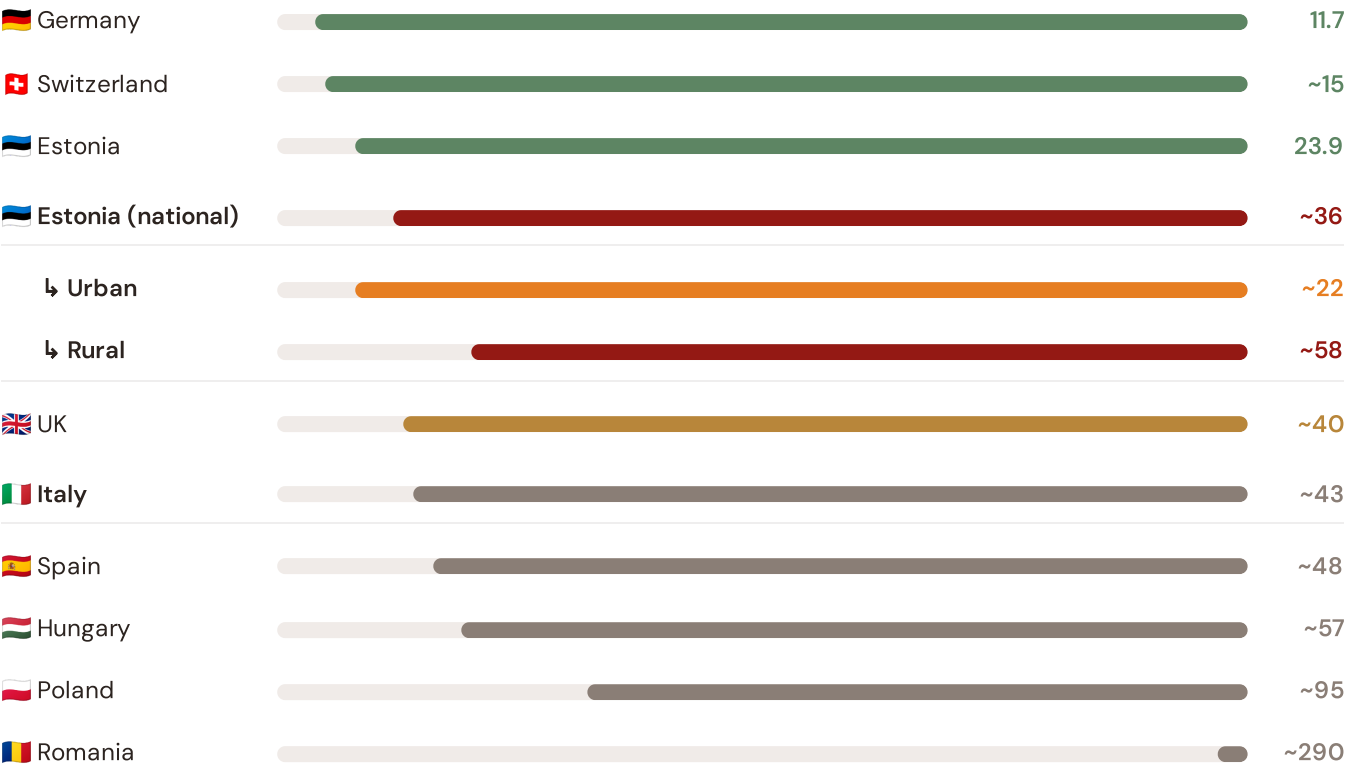
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the \$16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the NE-stressed cluster analysis hinges on Estonia's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



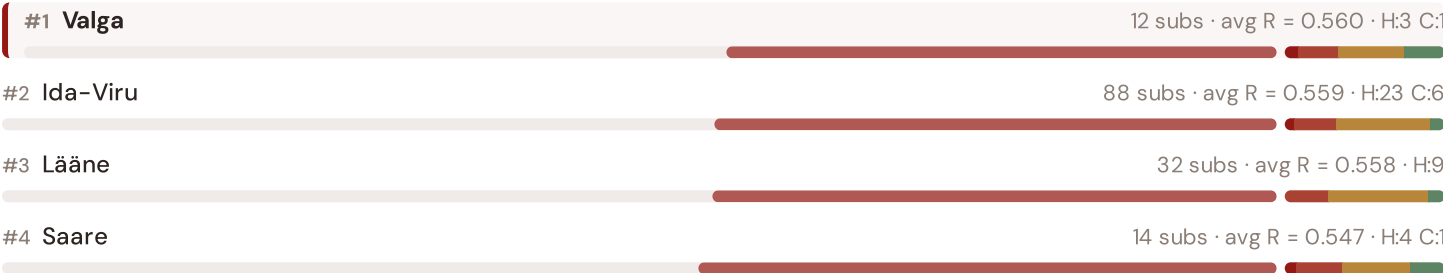
Source: CEER 7th Benchmarking Report (2022); ACM TIQE (2023); Bundesnetzagentur (2024). Estonia sub-categories from ACM territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

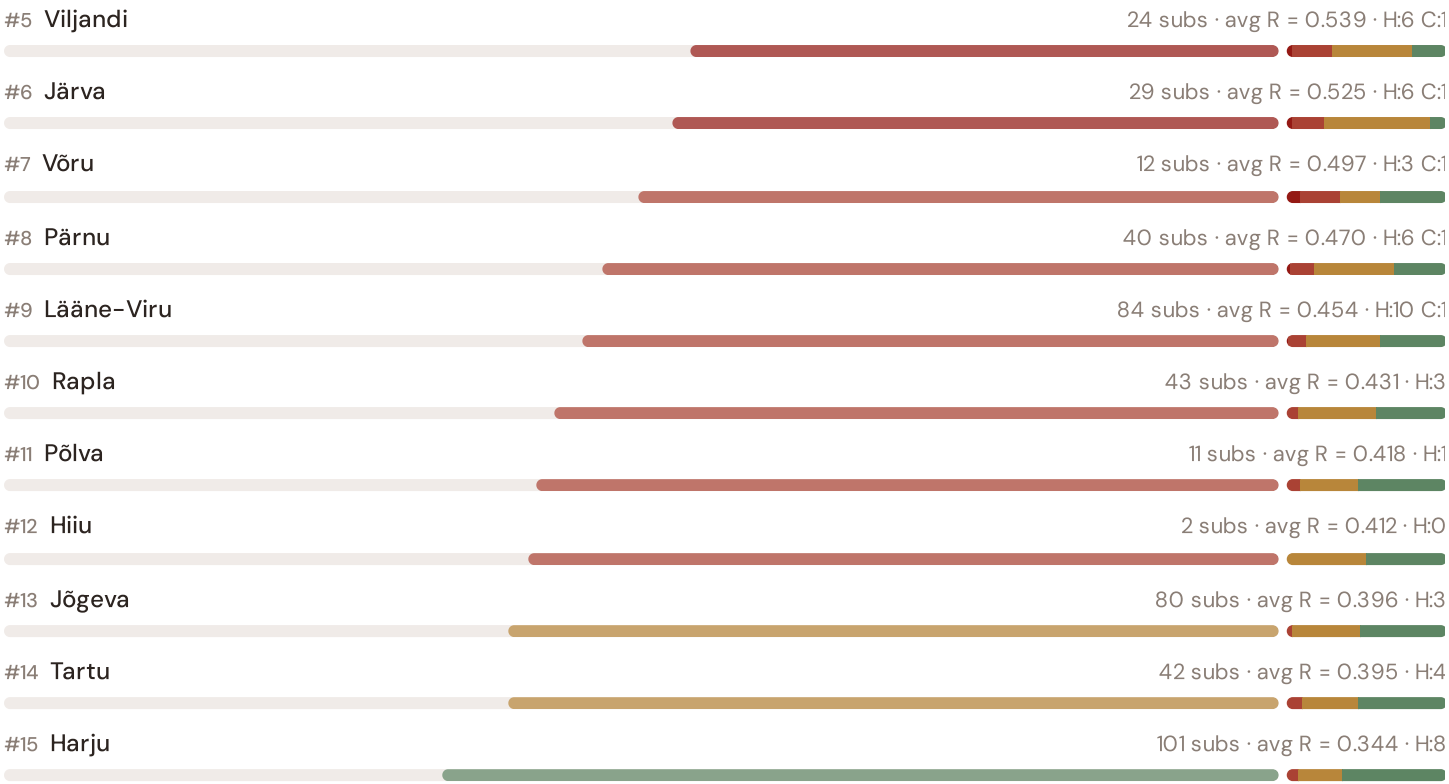
This month's key insight: The Valga corridor deep-dive shows **4 substations** (of 12) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.336 is **1.0× the fleet average** of 0.325. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 LU Maakonds

Where does the NE-stressed cluster sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





The Valga ranks **#1 of 15 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Valga, Ida-Viru, Lääne, while the three most resilient are Harju, Tartu, Jõgeva. 8 of 15 score above the fleet average of 0.455. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Elering/Elering and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3

§0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 614 substations with complete feature vectors. Average CI_width of 0.071 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 236 Low, 276 Medium, 89 High, 13 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 3

 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 3 Quarterly Sources

DESI, Maa-amet, Eesti Energia transition are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Eurostat NUTS-3** (18 variables → R3, E2).

FLEET SPOTLIGHT

Saare: Highest Risk Concentration

Saare has the highest concentration of High/Critical substations: **5 of 14 (36%)** are in the upper risk bands — avg R = 0.547. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Estonia utility, ACM, Elering, Elering, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 01 (Inaugural)

Pre-launch placeholder · First refresh 2026-07-09 · SSI Index v4.0.2 · 614 substations (LIVE OSM) · 4,757 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5

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